

Problem

Given that

$$f(t) = \sum_{n=\text{odd}}^{\infty} \left(\frac{20}{n^2 \pi^2} \cos 2nt - \frac{3}{n\pi} \sin 2nt \right)$$

plot the first five terms of the amplitude and phase spectra for the function.

Step-by-step solution

Step 1 of 1

$a_n = 20/(n2\pi^2), \quad b_n = -3/(n\pi), \quad \omega_n = 2n$

$$A_n = f(t) = \sum_{\substack{n=1 \\ n=odd}}^{\infty} \left(\frac{20}{n^2\pi^2} \cos 2nt - \frac{3}{n\pi} \sin 2nt \right)$$

$$= \frac{3}{n\pi} \sqrt{1 + \frac{44.44}{n^2\pi^2}}, \quad n = 1, 3, 5, 7, 9, \text{ etc.}$$

n	An
1	2.24
3	0.39
5	0.208
7	0.143
9	0.109

$\phi_n = \tan^{-1}(b_n/a_n) = \tan^{-1}\{[-3/(n\pi)][n2\pi^2/20]\} = \tan^{-1}(-n\times0.4712)$

n	ϕ_n
1	-25.23°
3	-54.73°
5	-67°
7	-73.14°
9	-76.74°
∞	-90°

